

MIKE SHE modeling of ecohydrological processes: Merits, applications, and challenges



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ABSTRACT

MIKE SHE is one of the most extensively used physically based distributed hydrological models that are capable of credibly interpreting the response mechanism of ecohydrological processes to ecosystem components and organization. However, implementing MIKE SHE in integrated catchment modeling is a complex task requiring an experienced modeling team with skills in a variety of disciplines, making it difficult for scientists without a background in hydrology or computer science to utilize this useful model. This review, from the perspective of ecology and environmental science, illustrates the important role of a physically based distributed model as a solution to issues related to ecohydrological processes, offering a major discussion of issues that received little attention from the previous reviewers and that would interest researchers without a hydrology background. The inevitable and possible difficulties confronting these potential users at the beginning and early stages of utilizing the model are also brought to light, with suggestions for solutions. This paper offers a brief review of MIKE SHE development and elaborates its main merits, key problems, and research on the model's theories and practices implemented by scholars. Lingering problems in current solutions are analyzed, and potential approaches to solutions are proposed. We hope that this paper provides some practical advice to researchers who want to initiate study of water issues related to modeling ecohydrological processes by employing MIKE SHE.

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1. Background of ecohydrological processes research

Because of progressively severe problems with water resources, strengthening research on water resources management and hydrological forecast has become increasingly urgent (Nalbantis et al., 2011). The purpose and focus of hydrology research have gradually progressed and shifted from simple flood prediction to comprehensive water resource management that requires more information on detailed hydrological processes (Jarsjo et al., 2012). Hydrological models were developed in order to deal with these issues, with the main type of hydrological model shifting from a rainfall-runoff statistical model, which is lumped and conceptual, such as the Stanford model (Crawford and Linsley, 1966), to a physically based distributed hydrological model (Refsgaard et al., 2010). Because lumped conceptual models cannot simulate hydrological process changes related to human activities (e.g., landscape changes), cannot efficiently solve problems of water pollution simulation at the watershed scale, and have difficulty providing an

integral simulation of the hydrological processes of surface water and groundwater, research on a physically based distributed hydrological model has been driven by both social need and computer technology development (Abbott et al., 1986a). Through 30 years of practices and research, that type of physical model has become the main approach for studying the response relationship between natural properties and hydrological processes (Arnone et al., 2011).

Ten years ago, the IAHS (International Association of Hydrological Sciences) proposed the first Scientific Decade plan (2003–2012), which focused on Predictions in Ungauged Basins (PUB) (Sivapalan et al., 2003). Currently, the theme of the new Scientific Decade 2013–2022 of IAHS has changed from PUB to Predictions under Change (PUC) related to water, earth, and biota in the Anthropocene, and its purpose is “to reach an improved interpretation of the processes governing the water cycle by focusing on their changing dynamics in connection with rapidly changing human systems (Montanari et al., 2013). With the present acceleration of human impacts, it is becoming increasingly clear that improved accounting for change, interactions, and feedback is necessary in order to provide a better interpretation of coupled human–natural systems (Jarsjo et al., 2012). The future of hydrology is closely related to our understanding of changing behaviors in the

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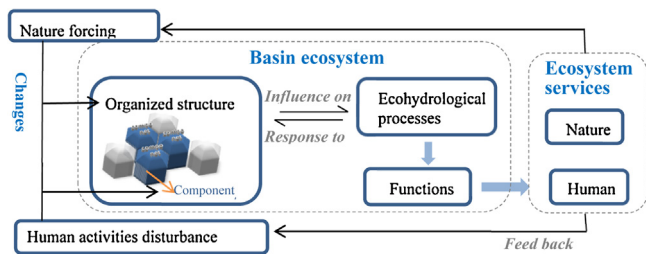


Fig. 1. Influence of changes in the components, structure, processes, and functions of the ecosystem with respect to hydrology.

interaction of hydrological systems and connected processes (Schaeffli et al., 2011). In particular, it is necessary to study and represent the connection between water and humans in greater depth, as this is one of the main drivers of change and is connected to both sustainable water use and sustainable development (Montanari et al., 2013). For instance, assessment of the hydrological influence of forest management measures is important for the ecosystem function of water conservation, while in the engineering practices of wetland restoration and recreation, predictions of the hydrological processes under environmental changes are essential in guaranteeing the engineering performance.

The change in objective from PUB to PUC indicates that the focus of hydrological research has changed from stationary to nonstationary – in other words, activities take place during the modeling for prediction purposes. Therefore, the models are unable to describe the reality of nature's characteristics and will no longer adapt for the new claims of PUC because they do not adjust the model parameters and structure according to the real changes in the components and organization in nature. As shown in Fig. 1, knowledge of the relationship between the components and the organization of the ecosystem and hydrological processes is the core and the key to quantitatively describing the hydrology resulting from system changes. Under this condition, there is increasing need for research in ecohydrological processes that can truly describe the components, organization, and process of the ecosystem and reflect the influence mechanism of the components and organization on the hydrological processes based on physical laws.

Even though papers have been using the term “ecohydrological processes” in recent years and although the number of results by “topic” in the “web of science” for research on “ecohydrological processes” is given as 265, few of those studies have offered a thorough exploration of the influence mechanism of ecosystem environmental factors on the hydrological processes that might allow consideration of spatial and temporal heterogeneity, due to the deficiency of the models employed. To respond to satisfy the new claims of research on ecohydrological processes, the MIKE SHE model is proposed here.

The MIKE SHE model was developed by the Danish Hydraulic Institute (DHI) on the basis of the SHE (Système Hydrologique Européen) code. It is a physically based, surface-subsurface integrated, unsaturated-saturated flow coupled distributed hydrological model and can simulate almost all important processes of the comprehensive hydrological cycle (Refsgaard et al., 2010). The complex model construction and sophisticated function of MIKE SHE inevitably lead to problems. As soon as the SHE code was first introduced publicly by Abbott et al. (1986a,b), they had already noticed the problems in the model and discussed the key issues after elaborating its advantageous features and envisaging future applications. Moreover, Beven (1989) responded to Abbott by raising a more fundamental criticism of the concepts and the methods of the SHE model, pointing out problems of overparameterization, equifinality, and uncertainty, and suggesting the use of the uncertainty analysis method in order to estimate model uncertainty.

Thereafter, the dispute about the SHE and MIKE SHE models centered mainly on these controversial views. Refsgaard et al. (2010) approved of the uncertainty analysis method, but also stated that “the scientific progress made during the past two decades on these fundamental issues has been very limited.” Regarding the necessity of uncertainty analysis in modeling, Ewen et al. (2012) held a view opposite to that of Refsgaard et al. (2010).

Today, the MIKE SHE model has already performed well in applications of ungauged basins, and the powerful capability of the physically distributed model will likely provide potential solutions and approaches for PUC and the research on ecohydrological processes in the future. It is one of the most extensively used physically based distributed hydrological models that is capable of credibly interpreting the response mechanism of ecohydrological processes to ecosystem components and organization. However, “to use MIKE SHE for integrated catchment modeling is a complex task requiring an experienced modeling team with skills in a variety of disciplines such as hydrogeology, soil science, agronomy and computational hydraulics, and such a multidisciplinary team does not often exist in a single research organization or public authority, where professionals are typically organized in mono-disciplinary departments” (Refsgaard et al., 2010); thus, it is hard to initiate research by implementing this useful model, especially for researchers with no background in hydrology or computer science. Therefore, a comprehensive understanding of MIKE SHE will be very helpful before charring out modeling practices, since targeted comprehensive introductions, such as reviews, are very limited. The earlier review by Refsgaard et al. (2010) is representative of the few, in which the key issues of the model are thoroughly and professionally expounded and discussed from the perspective of hydrology. This review, as a complementary elaboration from the perspective of ecology and environmental science, illustrates the important role of a physically based distributed model as a solution to issues related to ecohydrological processes, offering a major discussion of issues that received little attention from previous reviewers and that would be interest to interest researchers without a background in hydrology.

Due to the complexity of the model itself and the requirement of multidisciplinary, a comprehensive elaboration of its valuable features, key issues, and solutions may be helpful for researchers who are less familiar with the MIKE SHE model but are interested in employing a physically based distributed hydrological model to research about ecohydrological processes, especially those researchers without a background in hydrology science and computer science. The following offers a brief introduction to the development of MIKE SHE, summarizes the merits of its features, classifies its applications, and discusses its key problems as well as researchers' main efforts and attempts to solve these problems. A logical map of contents (Fig. 2) was drawn, in order to help in the understanding of the internal logical relationship between the features, applications, key problems and solutions, and to provide clear guidelines and convenient reference for readers who are likely to be interested in some special issues related to MIKE SHE. As shown in Fig. 2, MIKE SHE is capable of providing a good expression of the heterogeneity of natural properties and of providing detailed descriptions of physical hydrological processes; therefore it has been widely applied in the modeling of various catchments types, focusing on a variety of purposes, such as water resource management and hydrologic prediction under land use change and climate change. However, numerous parameters and a complex structure inevitably lead to overparameterization, as well as to problems of equifinality and simulation uncertainty. Attempts by scholars to solve these problems are analyzed, and potential approaches to solutions are proposed. Finally, by combining our understanding based on practical applications and the demands of the general trend in ecohydrological processes in

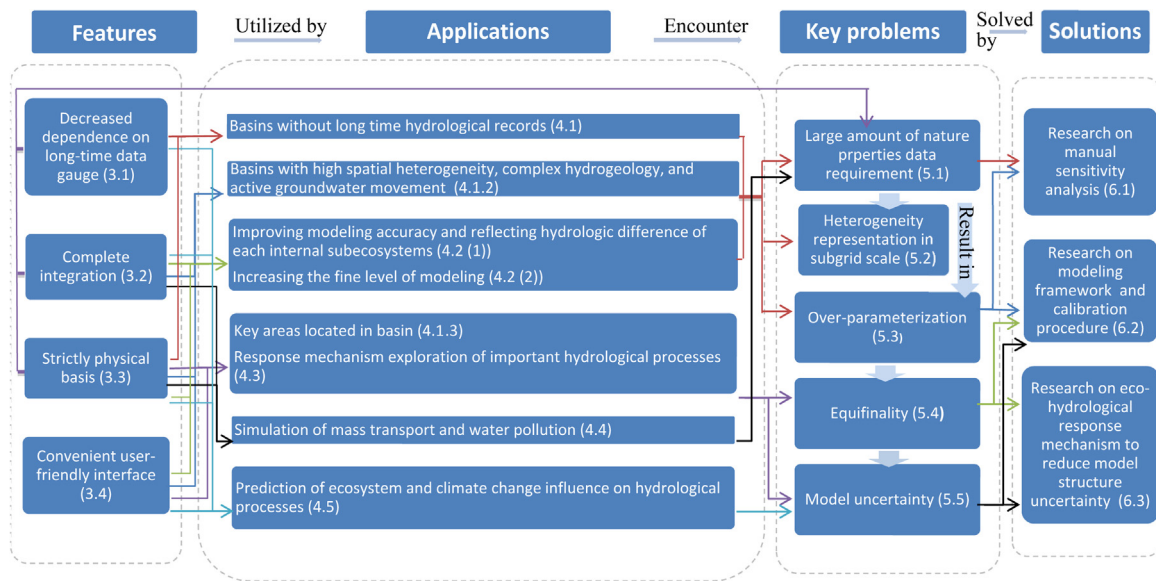


Fig. 2. The logical relationship map of the contents.

research on model capability, we present our perspective on the focus of future research. The inevitable and possible difficulties confronting potential users at the beginning and early stages of model implementation are also brought to light and solutions suggested. It is hoped that this paper provides some practical advice to researchers who want to initiate study of water issues related to modeling ecohydrological processes by employing MIKE SHE.

2. Introduction of MIKE SHE model

2.1. Structure of MIKE SHE model

The significant superiority of the physically based distributed hydrological model MIKE SHE, which integrates surface water and groundwater, is seen in three dimensions of the physical modeling of the main coupled and consecutive hydrological processes based on the laws of conservation of mass, energy, and momentum.

A spatial distribution of watershed parameters, climate variables, and hydrological response is presented through an orthogonal grid network and column of horizontal layers at each grid square in the horizontal and vertical, respectively. For advanced applications, MIKE SHE can simulate all the processes using physics-based methods. Meanwhile, MIKE SHE can also combine conceptual and physics-based methods-based on data availability and project needs (Graham and Butts, 2005). The detailed structure of the MIKE SHE code is illustrated in Fig. 3, and the hydrological processes and modules are listed in Table 1. The flexibility in MIKE SHE's process-based framework allows each process to be solved at its own relevant spatial and temporal scale.

2.2. Brief history of development of MIKE SHE

The first published application of the SHE code was the simulation of the 10.55 km² Wye River research catchment in Wales over several days (Bathurst, 1986). Because of computer limitations

Table 1
Main water movement modulators of the MIKE SHE code.

Modulars	Main hydrological processes	Numerical solutions	Solution type
Precipitation and evapotranspiration	Interception by the canopy Drainage from the canopy Evaporation from the canopy Uptake of water by plant roots and its transpiration Evaporation from ponded water Evaporation from the soil	Soil Vegetation Atmosphere Transfer (SVAT)	Physical
Unsaturated zone flow	Infiltration Moisture distribution Moisture deficit	Kristensen and Jensen Method Two-layer Water Balance Method 1-D Richards's equation, Finite Difference Method Gravity Flow Two-Layer Water Balance Lumped Unsaturated Zone Calculations Coupling to the Saturated Zone	Physical Experiential Physical Experiential Experiential Physical
Surface flow	Detention storage Surface runoff	2-D diffusive wave Saint Venant equations, Finite Difference Method Semi-distributed Overland Flow Coupling with MIKE11	Physical Experiential Physical
Channel flow	Flow and routing in rivers Flooding	1-D diffusive wave Saint Venant equations, Finite Difference Method	Physical
Saturated zone flow	Groundwater flow Groundwater exchange	3-D Boussinesq equation, Finite Difference method Linear Reservoir method	Physical Experiential

Based on Graham (Graham and Butts, 2005) and MIKE SHE Reference Guide (DHI, 2011).

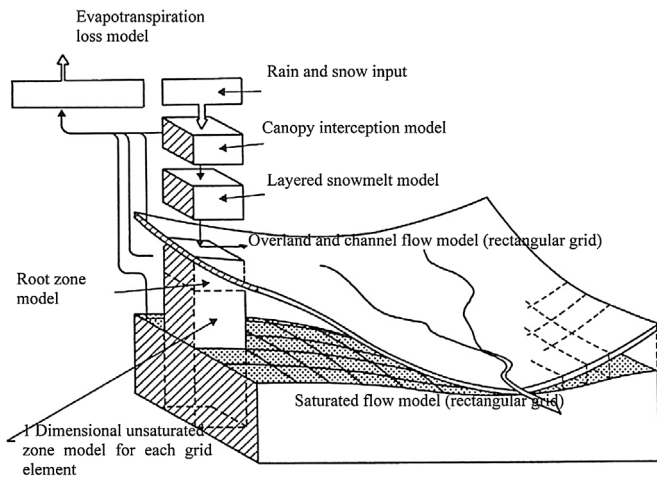


Fig. 3. MIKESHE model structure, reference (Refsgaard et al., 2010).

30 years ago, the model's ability to couple soil science, hydraulics, groundwater, and other physical processes was an important breakthrough. Before MIKE SHE was successfully employed to simulate the complete basin over a long period of time in India (Refsgaard et al., 1992), only Miles and Rushton (1983) had ever used an integrated physical distributed hydrological model for such a large-scale watershed simulation over a long period; however, its surface water model was lumped. Therefore, Graham and Butts (2005) stated that the major scientific contribution of the SHE and MIKE SHE model is having been the first actual successful application of a fully distributed physical model for a basin of more than 1000 km², and they considered it the representative integrated distributed physical model. The past 30 years of model development from SHE code to MIKE SHE code have been systematically summarized by Refsgaard et al. (2010) and divided into four stages.

The first stage is the period 1976–1986. The Association pour le SHE (ASHE), consisting of the British Institute of Hydrology, the Danish Hydraulic Institute (DHI), and the French consulting company SOGREAH, was established to develop the SHE model. The landmark event was the first publication of a small watershed hydrological simulation using SHE code (Bathurst, 1986).

The second stage, 1986–1992, is called the consolidation period. Compared with the 1986 code version, the most significant advancement in this stage was the coupling of different components. The major difficulty in this stage was finding a way to resolve the contradiction between the iteration of numerical calculation and the parallelism of different hydrological processes taking place simultaneously: e.g., the results of A process provide boundary conditions for interrelated B process, while the two processes take place in parallel. The contradiction was especially problematic in the coupling of the saturated zone (SZ) and the unsaturated zone (UZ), leading to various problems with the numerical algorithm (Storm, 1991). Due to the limitations of computer capabilities in this stage, it was difficult to execute larger-scale and more complex hydrological modeling. Each of the three partners made contributions in addressing these issues, although the major work was done by DHI. Model operation became more convenient, and DHI "ported all important improvements into one common code" (Refsgaard et al., 2010) at the end of their cooperation. After that, however, the three partners went their separate ways.

The third stage, 1992–2003, is the stage of soft packaging and dissemination, as well as the model transition period for SHE code to MIKE SHE code by DHI. This change ensured the investment of substantial resources and maintained the vitality of model. At the

end of the second stage, collaborators had recognized that with the requirements of practical applications and the adoption of a newer computer operating platform, a large number of updates to increasingly complex model codes needed to be maintained, so the securing of funding sources became an inescapable challenge. To that end, DHI adopted a new strategy to speed up the model packaging and actively promoted dissemination of the application in order to secure funding for future development.

The fourth stage, from 2003 to the present, is the integration period. Major progress is reflected in the delicate description of processes, convenient user-friendly interface, and integration with other model codes originating from different domains. Due to emphasis on environmental and climate change issues, demand for related research has increased significantly, making it possible to couple different models, even in different domains, to improve the MIKE SHE model. Suitable software techniques, such as the approaches of MODFLOW (Banta et al., 2008) or the more general OpenMI (Gregersen et al., 2007), were suggested by Refsgaard et al. (2010) to address compatibility issues in attempts to couple DHI models with other model software.

3. Superior features of MIKE SHE

As shown in Table 2, compared with other type hydrological models, the MIKE SHE model can better express the spatial heterogeneity of nature and is able to clearly describe complete integrated surface-subsurface hydrological processes on a physical basis. The favorable recognition and promotion of MIKE SHE is mainly due to its outstanding advantages. In general, compared with the currently widely used typical semi-distributed conceptual model SWAT whose computational unit is the hydrological response unit (HRU) (Arnold and Fohrer, 2005), MIKE SHE code computing on unit of grid is more feasible for expressing the continuous hydrological process from one region to its adjacent contact, while the processes are individual in each HRU of the semi-distributed model. In addition, MIKE SHE describes processes based on physical meaning while SWAT does so by empirical equation (especially soil water movement and the coupling of surface and groundwater), so that SWAT cannot show realistic water movement in environmental components (such as moisture content dynamics in each soil profile), whereas MIKE SHE can simulate the complete hydrological processes of integrated surface and groundwater. In terms of application domains, SWAT is not well suited to flat areas with a dense river network or regions where the groundwater is dominant, where MIKE SHE performed well. Despite constraints related to computer capability, the outstanding capabilities of MIKE SHE can be employed to explore the mechanisms of hydrological processes in response to the environment, rather than simply pursuing the improvement of the fit between simulation and observation. This superiority is generally recognized by hydrologic researchers. The main merits and demerits of MIKE SHE model are summarized in Table 2 in broad outline, and the four merits are described in detail here.

3.1. Independence from long-time data gauge

Unlike with a conceptual model requiring long-time data for calibration, the parameters required by MIKE SHE can be obtained directly from field measurement data for describing a basin's properties rather than relying on long-term monitoring of runoff data. This is extremely significant for ungauged regions (often the catchments of third-order tributaries as well as water resources regions) without long-term hydrologic records, where the MIKE SHE code makes hydrological modeling feasible, according to Abbott et al. (1986a).

Table 2

Hydrological model classification, application conditions, and advantages and disadvantages of MIKE SHE.

Model type	Mainly surface water	Groundwater	Integrated surface and groundwater	Advantages of MIKE SHE	Disadvantages of MIKE SHE
Distributed physically-based		HydroGeo-Sphere ^a FEFLOW ^b MODFLOW ^c	MIKE SHE ^d SHETRAN ^e	Independence on long-time data gauge	Numerous specific hydraulic properties are necessary
Semi-distributed physically-based	PRMS ^f		GSFLOW ^{g,1}	Complete integration of overland and ground water	Complex Spatial distribution of environmental factors are necessary
Distributed conceptual	TOP-MODEL ^h			Strictly physical basis	Complicated physical based algorithm of hydrological processes
Semi-distributed Conceptual	SWAT ⁱ HBV ^j			Convenient user-friendly interface	Strong nonlinear relationship between parameters and simulation results
Lumped conceptual	Stanford model ^k				

^a Brunner and Simmons (2012).^b Trefry and Muffels (2007).^c Prommer et al. (2003).^d Refsgaard et al. (2010).^e Bathurst et al. (2006).^f Markstrom et al. (2015).^g Markstrom et al. (2008).^h Page et al. (2007).ⁱ Arnold and Fohrer (2005).^j Engeland and Hisdal (2009).^k Crawford and Linsley (1966).¹ Coupling of MODFLOW and PRMS.

3.2. Complete integration

The MIKE SHE model dividing the integral nature system into discrete grids in three dimensions can reflect the heterogeneity of horizontal and vertical directions and can reasonably express the actual ecosystem characteristics. Additionally, it integrates surface and groundwater and can completely describe the main dynamical hydrological processes of the catchment, making it adaptable to regions of more complex hydrogeology. In addition, migration and transformation of pollutants and soil erosion can be simulated on the basis of physical water movement modeling (Abbott and Refsgaard, 1996; Mottes et al., 2014). These advantages have been verified in numerous practical applications in the past decades.

3.3. Strictly physical basis

As shown in Table 1, MIKE SHE, which solves the partial differential equations describing mass flow and momentum transfer, provides numerical solutions for surface water, groundwater, plant transpiration process, etc. based on physical mechanisms. In particular, the difficulty of combining an unsaturated zone and a saturated zone was addressed by employing sophisticated numerical solutions that couple different algorithms. MIKE SHE includes an iterative coupling procedure between the unsaturated zone and the saturated zone to compute the correct soil moisture and the water table dynamics in the lower part of the soil profile. Particularly in this part of the model, it is important to account for the variable specific yield above the water table, as the specific yield depends on the actual soil moisture profile and availability of that water (Graham and Butts, 2005). Thus, the model is capable of simulating more details of soil water movement such as water level, soil moisture content, and soil water tension. The coupling of unsaturated zone and saturated zone, which is not involved in the conceptual models, efficiently integrates the surface water and groundwater on a physical basis, allowing it to meet the requirements of simulating regions of more complicated hydrological processes (Christiaens and Feyen, 2001; Doummar et al., 2012;

Sørensen, 1996) and allowing for the solution of more comprehensive hydrological issues of water resources management. With respect to the hydrological modeling of landscape changes, the changed parameters of MIKE SHE, unlike in previous conceptual models, have clear physical meanings and detailed input and output clearly articulate the spatial heterogeneity and time continuity of different hydrological processes, making hydrological forecasts more realistic (Kalantari et al., 2014; Sultana and Coulibaly, 2011; Vansteenkiste et al., 2013; Wijesekara et al., 2014).

3.4. Convenient user-friendly interface

The transformation of MIKE SHE code from SHE code made user operation much easier and to some extent offset the adverse effects of operational barriers caused by complex model structures. As shown in Fig. 4, MIKE SHE code contains a good graphic interface, including multiple-choice modules of hydrological processes (Graham and Butts, 2005), efficient manipulation of huge input and output data, which can conveniently convert to and from EXCEL and ARCGIS files, and graphic visualization of simulation surface and groundwater movement (Kaiser-Hill, 2001), making MIKE SHE far more practical and superior for professional consulting. For example, the model was employed to simulate the complex interactive hydrological processes among different environmental components in the Danube lowland related to reservoir sedimentation, river infiltration, permeability between reservoirs and aquifers, agricultural water use, nitrate leaching affected by groundwater table, and river-reservoir operation (Refsgaard et al., 1998). Also, the South Florida Water Management District (SFWMD) (1996–present) employed MIKE SHE for the management system of flood control, water supply, and water environment quality (Yan et al., 1998; Long et al., 2015).

4. Extensive applications of MIKE SHE

Based on the above features, the MIKE SHE model has been widely employed in research on hydrology and water resources

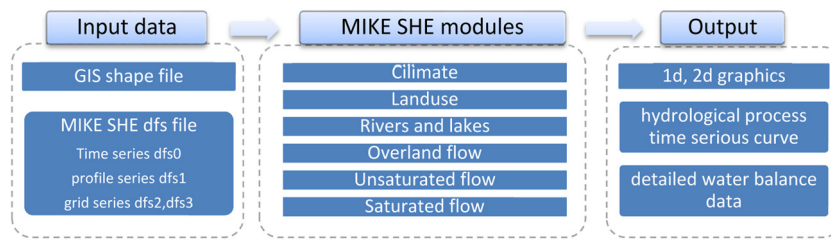


Fig. 4. Convenient user interface of MIKE SHE code.

for a variety of purposes. Surface water and groundwater have traditionally been managed separately, often in completely different government branches. It is now recognized that water resources problems cannot be treated in isolation, for the problems are seldom isolated and their solution requires a holistic approach to water management that must address different, often conflicting, demands for water. Wetland protection or the combined use of surface water and groundwater resources require integrated management of surface water and groundwater along with water chemistry and ecology (Graham and Butts, 2005), so numerous applications were classified by application areas by Graham and Butts (2005). Another most important purpose of the physical distributed model is to meet the requirements of modeling those ecohydrological processes that are subject to human activities and nonstationary climate such as land-use change, wetland management, groundwater extraction, irrigation-drainage and climate change, as well as the simulation of water quality and soil erosion (Refsgaard and Knudsen, 1996). As seen, MIKE SHE provides an approach to achieving goals not well met by conceptual models.

4.1. Applicability of various types of basins

4.1.1. Basins without long time hydrological records

The hydrological processes research on the small-scale basin is very important for explaining the hydrological phenomenon of the large-scale basin. Because the upstream hydrological processes taking place in the watersheds of third-order tributaries have a huge impact on the trunk river runoff, understanding the mechanism of the response of the hydrological process to the natural characteristics of tributary watersheds is the key to the explanation of the phenomenon of the trunk river.

Small-scale basins, whose areas range from dozens of square kilometers to hundreds of square kilometers, are relatively hydrologically independent catchments with a complete water cycle, and the detailed data of natural characteristics and hydrology are not as difficult to collect as for the large basin. Accordingly, deep exploration of the influence mechanism of the ecosystem on hydrological processes is more feasible. These small basins, located in the upstream of the trunk river and constituting the water source's conservation areas, are numerous and generally far from residential areas; thus, few of them receive attention from governments and researchers, and therefore most of them are lacking in long-term hydrology records. As a result, experiential models are not available in these areas because of the lack of long-term runoff data required for the calibration procedure. Whenever water resources management of a large river basin is urgently required, the hydrological processes of small catchments, which are basic hydrologic units, are needed. Because of the lack of long-term runoff data for the small basins, the physically based distributed models is the only available solution to simulate the processes, since a model based on field-measured data is generally better able to represent natural watershed properties and can efficiently improve the simulation accuracy of these ungauged regions (Windolf et al., 2011).

4.1.2. Basins with high spatial heterogeneity, complex hydrogeology, and active groundwater movement

This model offers a sophisticated integration of the surface and subsurface and is able to efficiently describe the complete hydrological processes. Hydrological processes in different environmental components vary significantly in regions of high heterogeneity, and the varieties can be described and simulated more exactly by the distributed model MIKE SHE, making this model suitable for regions with high heterogeneity. Especially in regions with complex vertical geological layers, a good coupling of SZ and UZ makes it possible to simulate the complex surface and groundwater movement, greatly expanding its application to catchments types, such as the active groundwater catchments of deserts, wetlands, volcanic regions, and karst regions. Many such experiments have been carried out, e.g., simulation of long-term agricultural canal irrigation and pumping irrigation to predict runoff, groundwater table and subsurface drainage in the North China Plain (Shu et al., 2012); research on the influence of ditch systems on the wetland water table in the Elmley Marshes (Thompson et al., 2004); stream-flow simulation of a volcano watershed in Hawaii (Sahoo et al., 2006); Gallusquelle spring flow simulation of a karst region watershed in southwestern Germany (Doummar et al., 2012); groundwater recharge and flood simulation of desert and arid area catchment in the Tarim Basin (Liu et al., 2007); hydrological cycle simulation of the Tarim Basin (Huang et al., 2010).

4.1.3. Key areas located in basin

Water movement is constrained by the watershed boundary, rather than the political boundaries, which always lead to fragments management activities, thus, problems sometimes need to be addressed at the local scale rather than basin scale (Graham and Butts, 2005). The basin ecosystem is composed of various types of subecosystems located in it, such as forest, wetland, grassland, farmland, and human settlement. These subecosystems are inextricably linked by the mass flow and energy flow because of the hydraulic connection between them, which integrates them into an indivisible unit. Therefore, when the focus region is one of the subecosystems, isolated research on hydrological processes inside the subsystem without consideration of the complex boundary conditions will not correspond to facts. Research on the local region must be established on the basis of whole basin hydrology modeling, which can provide boundary conditions. The discrete grids of the MIKE SHE model allows the specific local region to be defined arbitrarily by choosing the necessary grids according to purpose (Fig. 5a). Because the grid is the basic unit of numerical calculation, the calculation results of the selected grids composing the local region are also able to be extracted. In addition, intuitive graphs and numerous water flux data can be conveniently outputted by utilization of the user-friendly interface (Fig. 5b). The temporal sequences of recharge and discharge crossing boundary of the subecosystem can be calculated depending on the time-step volume accumulation of water balance (Eq. (1)). Through analyzing hydrological processes and water flux crossing the local region boundary,

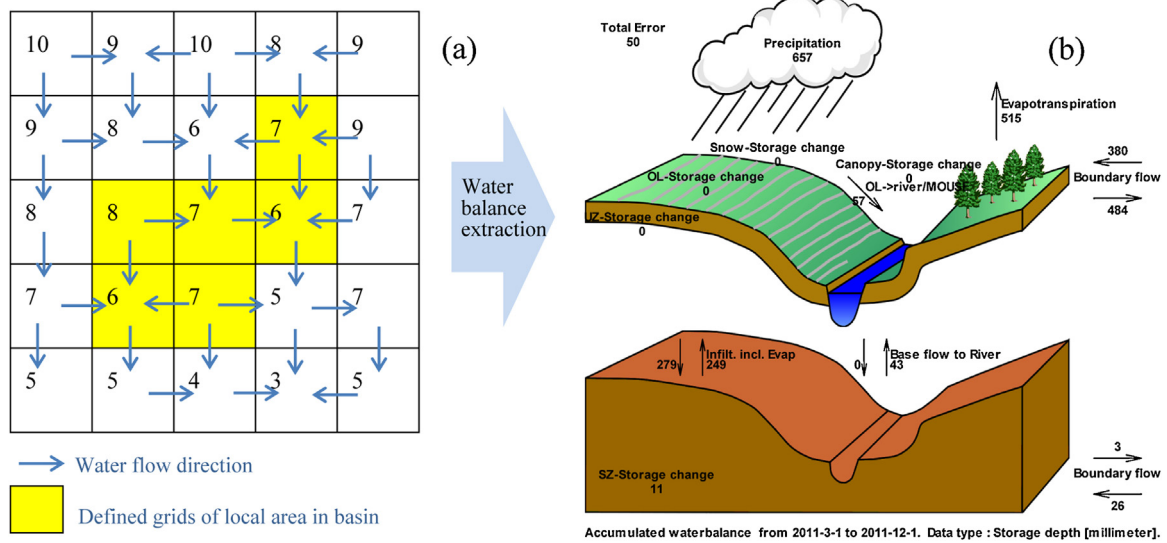


Fig. 5. Local area abstraction and water balance analysis.

we can have a clearer understanding of water movement in given region.

$$\begin{aligned}
 \text{Preci.} &= \text{Bou.Inf} - \text{SZSC} - \text{UZSC} = \text{EvapoT.} + \text{Riv.Runoff} + \text{Bou.OutF} \\
 \text{Riv.Runoff} &= \text{OL} - > \text{River} + \text{BaseF} \\
 \text{Bou.Inflow} &= \text{Bou.Sur.Inf} + \text{Bou.SZ.Inf} \\
 \text{Bou.Outflow} &= \text{Bou.Sur.Outf.} + \text{Bou.SZ.Outf} \\
 \text{SZ.Infilt.downward} - \text{SZSC} &= \text{SZ.infilt.upward} + \text{BF}
 \end{aligned}
 \quad (1)$$

4.2. Effective expression of spatial heterogeneity

4.2.1. Improving modeling accuracy and reflecting hydrologic difference of each internal subecosystem

For the river basin as an integral hydrologic system, there are both hydraulic relationships and differences between the internal different ecosystems. For the basin composed of significantly different subecosystems, regardless of differences during simulating, the model did not substantially reflect the actual natural conditions, in terms of a good fit between the simulation results and observation runoff data of the basin outlet. In order to improve the accuracy, the calibration procedure should include some test of the internal consistency of the distributed results from subcatchments. Together with a restricted use of calibration, a multisite, multiscale, and multicriteria validation is likely a good way to further constrain distributed models (Andersen et al., 2001; Moussa et al., 2007), which improve the internal runoff simulation accuracy depending on the fitting of simulation data of several points to their corresponding observational data, such as multisite calibration of runoff and groundwater table level in the Grote and the KleineGete basins of Belgium (Feyen et al., 2000); multisite calibration and validation approach tested on the Gardon catchment located in the mountainous Mediterranean zone of southern France using data gathered over a 10-year period for nine internal subcatchments (Moussa et al., 2007); and multisite calibration employing stream runoff data at three stations within the large Chaohe River basin in northern China (Wang et al., 2013).

4.2.2. Increasing the fine level of modeling

The size of grid is a key factor directly affecting the accuracy of model expression, which results in different simulation performances. The choice of grid size should be made based on the tradeoff of data acquisition capability and the minimum data

requirement for model accuracy. The influences of different grid sizes on simulation were analyzed in the Grote and the Kleine Gete basins, with results showing that within a certain range of grid size, simulation performances varied insignificantly, but beyond this size range, simulation results deteriorated significantly (Vázquez et al., 2002). In addition, the influences of different DEM (Digital Elevation Model) gridding methods on simulation were assessed using a grid size of 600 m (Vázquez and Feyen, 2007). Moreover, the superior capability of good spatial heterogeneity expression made it feasible to change spatial physical parameters according to realistic landscape changes in catchments, such as in the study of runoff response of semi-arid scrub region landscape changes caused by fire in California (McMichael and Hope, 2007).

4.3. Response mechanism exploration of important hydrological processes to eco-environmental factors

The accurate physical description and user-friendly interface contribute greatly to revealing the complex hydrological processes and exploring the response mechanism of processes to natural properties. Strict physical meanings and principles allow MIKE SHE to describe hydrological processes in great detail and clearly explain the specific hydrological processes and their interaction relationship as well as the chain reaction of hydrological response after parameter changes. MIKE SHE provides approaches to revealing and exploring the invisible inner hydrological processes and to assessing the region's hydrological characteristics depending on observation of runoff, groundwater levels, soil moisture, and other hydrological phenomena. For example, Doummar et al. (2012) and Sahoo et al. (2006) individually found that saturated flow plays the dominant role in karst regions (the Gallusquelle spring catchment, located in the Swabian Alps in Southern Germany) and volcanic geology regions (the combined Manoa–Palolo watershed, located between the Koolau Range and Mamala Bay, Hawaii), which were mainly affected by the horizontal saturated hydraulic conductivity of geological layers. Feyen et al. (2000) also confirmed that saturated flow had a great influence on hydrological simulation results in the region of the Gete catchment located to the east of Brussels, Belgium, with its fine permeable loamy sand surface soil, while in the farm region of the North China Plain, the main influencing factor on groundwater in the short term was pumping irrigation (Shu et al., 2012).

4.4. Simulation of mass transport and water pollution

Because of its detailed description of hydrological processes and physical approaches, mass transport and water quality modeling, based on water movement, is regarded as another important advantage of the distributed physical model MIKE SHE. For example, it was applied to nutrient modeling at basin scale (Styczen and Storm, 1993), in pesticide simulation (Thorsen et al., 1998), nitrate leaching simulation (Refsgaard et al., 1999), physically based integrated modeling of nitrogen cycle in catchment scale (Hansen et al., 2009), nitrate reduction simulation in a complex geologically heterogeneous region (Refsgaard et al., 2014), phosphorus transport simulation in Everglades National Park (Long et al., 2015), and agrochemical transport simulation in an agriculture area (Kourgialas and Karatzas, 2015). However, compared to a large number of water movement simulations, the number of mass transport and water quality simulations is relatively smaller; this shows that the research and practices of fundamental water movement modeling have not being so numerous and proficient as to sufficiently sustain further research on water quality simulation. Therefore, the theory and practice of the model still need further improvement.

4.5. Prediction of ecosystem and climate change influence on hydrological processes

Prediction of hydrological processes under environmental changes is another important application of the MIKE SHE model. Climate changes directly impact precipitation and evapotranspiration. Experiential models based on statistics of historical rainfall and runoff data cannot lead to a confident prediction in cases where the climate has changed beyond the range of historical conditions. In a study to evaluate the impact of long-term climate change on the runoff of the rivers in Latvia (Bethers and Sennikovs, 2009), regional climate models (RCMs) accessible via the European Commission research project “PRUDENCE” EVK2-CT2001-00132 were used to predict the future climate. Comparison of RCM calculations for the reference (or control) time period 1961–1990 and for the climate change scenario A2 for time period 2071–2100 allows ensemble evaluation of run-off change forecast for the territory of Latvia. The downscaling procedure was employed for temperature, precipitation, and relative humidity data series for the contemporary climate (1961–1990) and climate change scenario A2 (2071–2100).

In addition, with respect to conceptual models based on the relationship of experiential statistics for independent variable and dependent variable, when the relationship changed due to environmental change, the experiential model actually had already transformed into a new one with a relationship different from previous one. In that case, physically based models, which are independent of historical statistics and are able to change the model depending on modification of the specific parameters according to changes in the environmental factors, are necessary in order to provide a more confident prediction. Consequently, predictions employing the MIKE SHE model have become more and more popular in recent years, as in studies of the impact of past and future land-use changes on hydrological processes related to the complex surface-groundwater interactions in the watershed of the Elbow River, covering an area of 1238 km in Alberta (Wijesekara et al., 2014); a simulation study demonstrating how the composition and configuration of land use measures affect discharge at the catchment outlet differently in response to storms of different sizes (Kalantari et al., 2014); an assessment of the respective impacts of land use change and climate change on decadal streamflow through use of three complementary models (Wang et al., 2013); a prediction of land use and climate change influence on groundwater and ecosystems at the middle reaches of the Tarim River (Keilholz et al., 2015).

Furthermore, the uncertainty of modeling results under the condition of climate and land use change should be considered. Any model is an abstraction, simplification and interpretation of reality. The incompleteness of a model structure and the mismatch between the real causal structure of a system and the assumed causal structure as represented in a model always result in uncertainty about model predictions. The considerable challenge faced in model applications is that predictions are required beyond the range of available observations, either in time or in space, e.g. to make extrapolations toward unobservable futures or to make predictions for natural systems that are likely to undergo structural changes (Refsgaard et al., 2006). Therefore, in order to reduce the adverse impact on prediction results induced by uncertainty, consideration of the issues of model uncertainty is strongly suggested. For example, in order to reduce the uncertainty due to model structure, ensemble combinations modeling encompassing 10 models (contain MIKE SHE) were employed to assess the impact of land use change on hydrology (Viney et al., 2009). The best model combination methods are a trimmed mean (constructed from the central four or six predictions each day) and a weighted mean ensemble (with weights calculated from calibration performance) that places relatively large weights on the better performing models. Furthermore, two distributed models of MIKE SHE and WetSpa were compared for their estimation of climate change impact on flow regimes in a medium-sized catchment in Belgium and it was found that the model structural uncertainties for high flow predictions were rather limited, but the projected low flow changes differed significantly over the models and even exceeded the uncertainty by the expected climate trends (Vansteenkiste et al., 2013).

5. Key problems and improvements

Despite the obvious advantages of the distributed physical model, Abbott et al. (1986a) had foreseen, when it first appeared, that the SHE model would face major difficulties, such as: (a) the requirement of multiple kinds of expertise for both developer and practitioner; (b) computing capability and the constant need to keep the model's computational burden adapted to current main-frame computers; (c) the need for potentially large amounts of data. Beven (1989) offered more fundamental critiques in response to Abbott et al. (1986a). We summarize the key problems of his critiques in the following 4 points (reference to the views of Refsgaard et al., 2010): (1) Model structure uncertainty: The model is an extreme simplification of reality where equations cannot guarantee description of the actual complex three-dimensional spatially heterogeneous and time varying system. (2) High degree of uncertainty in estimating the grid square values: The inputted grid values of distributed model are “a priori”; there must therefore be problems of how to choose the representative sites that can reflect the heterogeneity, and of what measurement techniques might be appropriate to obtain the required effective grid-scale parameters. (3) Spatial heterogeneity: At subgrid scale, spatial heterogeneity was not included in the physically based models, and using grid square effective parameter values implicitly assumes that a grid square is homogenous. Therefore, in the sense of grid scale, the model remains a lumped model at a more sophisticated level. (4) Predictive uncertainty: “Problems associated with errors in model structure, estimation of parameter values and the specification of initial and boundary conditions must result in a significant degree of uncertainty associated with the model predictions” (Beven, 1989). Especially in circumstances of predicting the effects of land use changes, the uncertainty is fatal to predictive credibility, because no calibration of parameter values by comparison with observed data is possible.

The main reasons for these problems can be summarized as follows: Numerous parameters are required to express heterogeneity; contradictions are generated between finite available data and infinite requirement; and a huge number of parameters leads to overparameterization and then results in equifinality of modeling, which furthermore causes difficulties for identification and selection of parameter sets and then leads to uncertainty of modeling. Confronted with these problems, researchers have made many efforts to find solutions for them.

5.1. Data requirement

Detailed types and spatial variability expression of parameters make the distributed physical model outstanding for fine description of natural properties, but also dramatically increase the data quantity (Feyen et al., 2000). The required data involve various environmental factors affecting hydrological processes, and should represent the spatial variability of environment, so the difficulty is not only the requirement of a huge amount of data, but also the concern of whether the heterogeneity can be efficiently expressed by survey data. Unlike other types models, such as SWAT, the MIKE SHE model is better suited to basins of hundreds to thousands of square kilometers, and requires more accuracy of environmental spatial distributed data, which usually cannot be well met by the universal database of governments, and is more dependent on specialized research and field measurements of the modeling area, especially in the water source region of rivers upstream, which usually lack precise information and detailed field surveys. Thus, the data requirement of the MIKE SHE model increases the number and difficulty of investigations, which in turn increases the survey burden and the subsequent risk of waste of human resources and finances (Feyen et al., 2000).

Current users generally recognize that the number of calibrations of the free parameters should be as few as possible, in order to ensure that the observed data can provide sufficient information to make parameter estimation (Hill et al., 1998). The first application of SHE (Bathurst, 1986) allows the changing of parameters causally to fit the objectively observed data during the calibration, a point criticized by Beven (1989) as overparameterization. In the next major SHE study (Beven and Binley, 1992), a more rigorous approach, estimating the representative parameter values depending on corresponding soil types, vegetation types, geological layers etc., was developed to reduce the number of free parameters in the subsequent calibration. However, 26 free parameters were still generated after using this approach in the simulation of an 820-km² basin in India (Beven and Binley, 1992). In the simulation of the Seine basin (with fewer available basin parameters than in India), the free parameters were decreased to only four (Andersen et al., 2001).

5.2. Heterogeneity representation

Like the semi-distributed model, the distributed physical model, which calculates based on the unit of grid, also cannot reflect the heterogeneity within a computational cell. Beven (1989) called this subgrid scale heterogeneity, believing that the influence of the ignored heterogeneity in subgrid scale on simulation results would barely be assessed. The problem of spatial heterogeneity expression and the lack of heterogeneity in subgrid scale are the major flaws of the distributed physical model (Hansen et al., 2007, 2008). Seyfried and Wilcox (1995) discussed the importance of spatial variability related to process description, grid scale, and measurement scale, and concluded that when using a physical model, it is necessary to include the assessment of natural characteristic expression by model. In fact, the prerequisite of ensuring the spatial variability

superiority of the distributed model is to make certain that spatial variability can be authentically described.

The natural characteristics are expressed by the digitization parameters of numerical modeling, but the full realistic details of natural characteristics cannot be described by the parameters; thus, the real heterogeneous environment, within a certain range, must be described by fictitious homogeneous numerical parameters, and the key is whether the value of the substituted fictitious parameters can reflect the general characteristics of the environment. We suggest that ecological theory and the method of quadrat survey can be employed to deal with the problem of heterogeneity expression to some extent. Regions with similar properties can be seen as one sample area whose general characteristic can be indicated by the average characteristic of quadrats, which means that the general characteristic of sampling area is presented by virtual values averaged by quadrats. In the specific sampling area, the variability assessment of quadrats can be applied to indicate the heterogeneous degree of this region, and the empirical method can be used to determine the threshold value of heterogeneity, so that the region whose assessment value is lower than threshold can be regarded as homogenous. The method can efficiently reduce the quantity of parameters under the condition of less influence on actual heterogeneity expression.

5.3. Overparameterization

Overparameterization is caused by both the many kinds of parameters and the great amount of spatial heterogeneity in describing data. On the one hand, overparameterization causes the core dominant parameters affecting the main hydrological processes to be hidden in the huge number of parameters, and to be barely recognized, which weakens the physical meaning of the parameters. Moreover, the time-consuming and numerous parameters for calibration also detract from the superiority of MIKE SHE. On the other hand, users can be confused about where to begin when faced with so many parameters and model components at the beginning of modeling, increasing the difficulty of use. Therefore, Beven (1989, 1996) considered these distributed physical models to still be lumped conceptual models with an increased number of parameters that can be changed during the calibration process because of overparameterization. Grayson et al. (1992) even doubted whether there were any substantial differences between the distributed model and lumped conceptual model, other than the increased number of calibration parameters.

Only models with good operability can be easily disseminated and widely applied. In the application of any model, the first difficulty users encounter is the blindness of parameter selection. The requirement of mass parameters of MIKE SHE undoubtedly raises the threshold for users. And to meet the mass data requirement, the choice of parameters and modules is also blind. Therefore, to reduce the blindness of free parameter selection and the risk of erroneous judgments for key parameters, the environmental characteristics, main hydrological process, and dominant sensitive parameters of various specific types of basins should be gathered and classified, to formulate different types of parameter sets corresponding to specific basins that can be used as guides. As Refsgaard and Henriksen (2004) wrote, "Statements on model capabilities can therefore not be made universally but should be conditioned on specific sites and specific types of applications." Modeling applications must depend on the specific conditions of the basin and should be guided by them.

5.4. Equifinality

The phenomenon of equifinality is commonly found in a variety of hydrological models (Schoups et al., 2008) and means that

different parameter sets can all achieve good fit with the observed data, while the fatal problem is that the set reflecting actual natural characteristics cannot be confirmed (Beven, 2001, 2006). The conceptual model parameters do not have clear physical meanings, as the processes of parameter calibration are often a series of attempts of random permutation and combination, which inevitably leads to equifinality, while the main reason for the consequences of the distributed physical model is overparameterization. On the one hand, the huge amount of data confusing the recognition of key parameters and the physical meaning of parameters gradually disappear during the casual adjustment of calibration intended to achieve a good fit. On the other hand, when there is no exact recognition of the most important hydrological process and the corresponding dominant parameters, the effect of hydrological process in response to key parameters will be weakened due to the adjustment of other parameters. If the key parameters cannot be grasped in the process of parameter adjustment, the calibration process can likely become a “mathematical game.”

5.5. Model uncertainty

Model uncertainty is the most serious consequence of equifinality because of the barely distinguishable good-fitting parameter sets, which in particular give rise to problems of model prediction (Beven, 1989). Various good-fitting parameter sets often can be generated during the modeling, while actual characteristics of the natural basin are relatively stationary; therefore, not all good-fitting parameter sets are feasible for truly representing the natural basin. Consequently, the hydrological model established under this condition is uncertain. Model validation can provide credible evidence only for the existing conditions, but there are no guarantees in the prediction of conditions beyond the existing validation. Statements of validation methodology also indicate the existence of uncertainty. For example, Refsgaard et al. (2010) refer to the “principle that a model should never be considered universally validated, but can only be conditionally validated restricted by the availability of data and specifically performed validation tests,” while Lane and Richards (2001) also argue that “evidence of a successful prediction in observed spaces and times (conventional validation) cannot provide a sufficient basis for use of a model beyond the set of situations for which the model has been empirically tested.” The purpose of validation is to ensure that the model contains every environmental condition that can possibly exist (e.g., arid climate, humid climate, extreme weather, etc.). The main hydrological processes and dominant environmental factors occurring in different conditions are not the same; thus, if some important processes did not take place during calibration and validation, the dominant key parameters of these hydrological processes will very likely not receive enough consideration. In the view of Abbott et al. (1986a), one important mission of the physical distributed model is to solve the problem of hydrological modeling for prediction under climate and environment changes. Therefore, considering the model uncertainty in hydrological prediction under land use change and climate change is very necessary.

With respect to the problem of uncertainty, Beven (1989, 2001) and Beven and Binley (2014) believed that the defect of the model structure’s uncertainty cannot be substantially resolved, so he proposed uncertainty analysis methods that can assess the simulation uncertainty to deal with the problem, and developed the GLUE method to quantify the assessment. GLUE has been applied in much of the practical simulation research of MIKE SHE (Christiaens and Feyen, 2002; McMichael and Hope, 2007; McMichael et al., 2006; Blasone et al., 2008; Vázquez et al., 2008, 2009). Currently, research on uncertainty analysis has significantly increased (Refsgaard and Henriksen, 2004; Rosbjerg and Madsen, 2005). Pappenberger and Beven (2006) have furthermore emphasized that “uncertainty

assessment is today a mandatory element of a good modeling practice.” In order to better meet that need, MIKE SHE provides several alternative physical and conceptual approaches for each component of the model (Butts et al., 2004).

Beven (2002) proposed the alternative blueprint of the physical distributed model, including multiple contending model structures and a variety of parameter sets within each model structure, which was subsequently suggested by Refsgaard et al. (2010) as a pragmatic and realistic way forward, and is also strongly advocated by Neuman (2003), Poeter and Anderson (2005) and Refsgaard et al. (2006). Such a strategy implies a need among practitioners for model codes with alternative process formulations and software platforms, including several model codes and uncertainty assessment tools, so that a modeler can easily construct a suite of models with alternative model structures and uses these to assess prediction uncertainties. Kumar et al. (2015) chose eight popular hydrological models, including MIKE SHE, to form several sets of multi-model combinations for simulating river discharge, and then made the uncertainty analysis.

Refsgaard et al. (2010) stated: “We do not believe that the fundamental problems related to imperfect model structures and spatial heterogeneity at subgrid scale can be solved in a foreseeable future”, because researchers cannot acquire enough data to describe all the details of the geological zones and ecosystems. This opinion implied a pessimistic perspective with respect to the solutions of fundamental problems. Although the uncertainty analysis method proposed by Beven (1989, 2001) has generally acquired the approval and support of many scholars and has gradually come into the mainstream in hydrological modeling research, some scholars still hold different views. Ewen et al. (2012) and Refsgaard et al. (2012) have discussed the importance of uncertainty analysis, with the latter insisting upon the necessity of uncertainty analysis in modeling research, and the former arguing that uncertainty analysis is not the most important element, but that more efforts should target “testing fitness for purpose.” It is worth noting that the superiority of MIKE SHE, as well as the ongoing scientific and philosophical arguments and discussions about the model, will continue to make a positive contribution to the development of hydrological models and possibly will provide new concepts and approaches for solving model uncertainty problems.

6. Conclusion: vision for future research

Although there are still many problems to be solved, MIKE SHE does hold a special appeal, with its good potential for extensive utilization in exploring the mechanism of ecohydrological processes in future research. We believe that with regard to the issues of uncertainty problems, hydrological researchers have been overly concerned with reliance on uncertainty analysis methods to evaluate the uncertainty, while they have ignored the strong capabilities of the model itself for physically modeling hydrological processes to reduce the possibility of uncertainty. Therefore, a more rigorous calibration framework with a logistical procedure needs be developed.

Additionally, physical modeling should be further explored as a means of revealing natural internal hydrological processes. MIKE SHE provides the tool and approach to modeling the physical processes of natural hydrology, and it can be employed to quantitatively describe the main hydrological processes taking place in the study area. The specific detail mechanism of the influence of key ecological factors on hydrological processes can be recognized and explored by analyzing the specific ecohydrological process change in response to the corresponding parameter change and the chain reaction of interrelated hydrological processes. Basically, this is also an ecological model that elaborately illustrates the relationship

between the natural hydrological dynamics and environmental factors on the watershed scale. By analyzing the digitization principles of parameters and numerical solution approaches of processes, the modeling environment can be further compared with the recognized realities of the natural characteristics, determining the parameters that are closer to the natural reality. Based on the above analysis, we propose that future research should focus on the issues described below.

6.1. Research on manual sensitivity analysis

Reducing the number of free parameters is particularly important for the distributed physical model with numerous parameters. Sensitivity analysis is widely used in various hydrological models to reasonably reduce the number of free parameters and improve simulation efficiency, which utilizes statistical criteria (such as residual errors, efficiency coefficient, etc.) that qualitatively evaluate the differences between simulated and observed values, to rank the sensitivity hierarchy of various parameters. The series of parameters is always generated from a single factor value gradient with a definite interval. In other words, it is a series of parameter sets with only one changing parameter, while the others remain unchanged. The method effectively reduces the disturbance of insensitive parameters in the calibration process. Automatic calibration methods such as those using Latin Hypercube Sampling are always being employed in the sensitivity analysis in attempts to reduce insensitivity parameters in conceptual hydrological models (Cheviron et al., 2011; Huang et al., 2014), but the methods are difficult to use with efficiency in current distributed physical models because of the computational burden beyond the current generation mainframe computers: e.g., in MIKE SHE, the 3D structure formed with grids makes a one-year simulation of several hundred square kilometer catchments with 300-m grid size, taking hours in a current mainframe computer with CPU of 3.4 GHz. Obviously, automatic calibration of this type of common distributed modeling, which requires simulations over several thousand times, is scarcely being carried out at present. Consequently, even though sensitivity analysis has already been widely employed in the modeling application of MIKE SHE in the past 20 years, automatic calibration is still not suitable for the distributed physical model in making the sensitivity analysis, and a reasonably efficient manual calibration method is urgently needed.

6.2. Research on modeling framework and calibration procedure

Faced with the problems of overparameterization and equifinality, a systematic and rigorous modeling framework and logistical calibration procedure are suggested to reduce model uncertainty caused by mistakes of operation (Refsgaard, 1997; Feyen et al., 2000; Moriasi et al., 2007). The primary framework requirement is for the model user to have a comprehensive and profound understanding of the actual situation of the study area, just as Refsgaard et al. (2010) point out: “Due to the multidisciplinary complexity of integrated catchment modeling, the requirements of modeling team expertise are higher than for simpler types of modeling.” The simulation performance using distributed hydrological model in fact reflects the modeler’s degree of recognition of the region. Based on that understanding, a rigorous logistical calibration procedure can efficiently reduce the uncertainty caused through manual operation.

The application of MIKE SHE carried out in the hydrological modeling of the karst region (Gallusquelle spring catchment, located in the Swabian Alps in Southern Germany) has a relatively complete rigorous framework (Doutmar et al., 2012): in that study, (1) first, watershed characteristics were described in detail; (2) next, during the calibration process, sensitivity analysis of a series

of parameters with definite value interval was made to rank the hierarchy of parameter sensitivity, and then the parameters significantly affecting the simulation were selected according to their sensitivity; and (3) finally, the changed water balance time series curves of important hydrological processes after parameter adjustment were individually compared with corresponding previous unchanged ones, in order to reveal and analyze the mechanism of specific parameters’ influence on the corresponding hydrological processes. However, Doutmar et al. (2012) did not elaborate upon the calibration details for adjusting the parameters and the way to gradually increase the degree of fitting with the observed data. The exploration of calibration procedure and method is very useful and valuable in making the modeling operation more feasible for users; thus, more attention should be paid to it in future improvements of MIKE SHE.

6.3. Research on the response mechanism of ecohydrological processes to eco-environmental factors

In their conclusion on model development trends, Refsgaard et al. (2010) suggested that uncertainty analysis was still the main way to solve the uncertainty problem and that it should continue to be developed and extended. We agree with the necessity of continuing to develop uncertainty analysis methods, but we also find that in the research of the past forty decades, too much attention has been paid to the post-treatment as uncertainty analysis (although this is important), while too rarely have efforts been made and exploration undertaken in skillfully modeling detailed hydrological processes. Refsgaard et al. (2010) even pointed out that “the dissemination of the technology (MIKE SHE) is still proceeding today with a speed that is much lower than its potential,” implying that the potential capability of MIKE SHE was not being given full play, with the main constraints being institutional. They added that “to use MIKE SHE for integrated catchment modeling is a complex task requiring an experienced modeling team with skills in a variety of disciplines” and that the promotion of executable modeling operation approaches and skills should be paid more attention in future study.

Concerning the applications and developments of MIKE SHE, there has long been insufficient research on the potential capability of the MIKE SHE model itself, and we consequently argue that research should focus on the capability of using the model to recognize and explore hydrologic characteristics and detailed ecohydrological processes of specific regions. By means of detailed ecohydrological process analysis, the modeler can reveal the mechanism of physical hydrological processes in response to natural properties of eco-environmental factors and then interpret the physical meanings of the specific modeling structures and parameter sets. In this way, there would be reduced uncertainty during simulation and calibration depending on a comparison between model parameters and natural characteristics, as well as simulation results and actual hydrological phenomena. Consequently credibility of prediction would increase.

Conflicts of interest

The authors declare no conflict of interest.

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